
What is Online Privacy Worth?

Evidence from an Information Provision Experiment

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Privacy regulation and information frictions

High concern, limited knowledge (Pew 2023)

- **81%** are concerned about how companies use collected data
- **67%** understand little or nothing about what companies do with their data

Policy response (e.g., GDPR)

- Clearer information disclosure on privacy practices
- Additional consumer control over data

Evaluation challenge

- Firm-side costs are well documented
- Consumer benefits are hard to quantify

Research questions and contributions

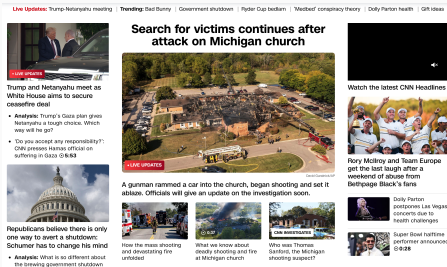
Research questions:

1. **What is online privacy worth?**
2. **Can privacy information align beliefs with reality—and change behavior?**

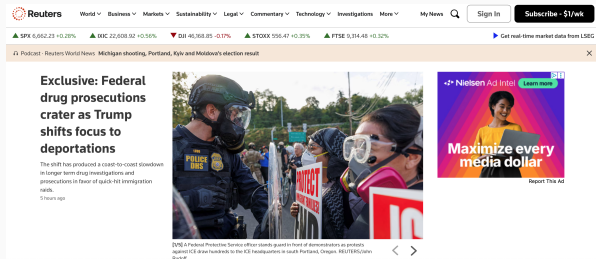
Contributions:

- **Define and construct a website privacy attribute space** — build a new dataset of 19 Collection / Use / Control privacy practices for the top 500 websites
- **Estimate consumer valuations for privacy practices** — conjoint choices recover dollar-valued willingness to pay for each privacy attribute
- **Build a Chrome extension** — delivers interventions and measures real browsing data
- **Identify disclosure mechanisms** — separate search costs from complexity in a randomized field experiment

Stylized model of website choice



(a) CNN



(b) Reuters

Why might a user choose website A over website B?

$$u_{ij} = \text{Direct utility}_{ij} + \text{Privacy}_{ij}$$

- **Direct utility:** content quality, convenience, habit
- **Privacy:** data practices differ across websites

Unpacking consumer privacy online

$$\text{Privacy}_{ij} = \sum_{k=1}^{19} \beta_{ik} \theta_{ijk}^b$$

β_{ik} : how much user i **values** attribute k · θ_{ijk}^b : i 's **belief** that site j has attribute k

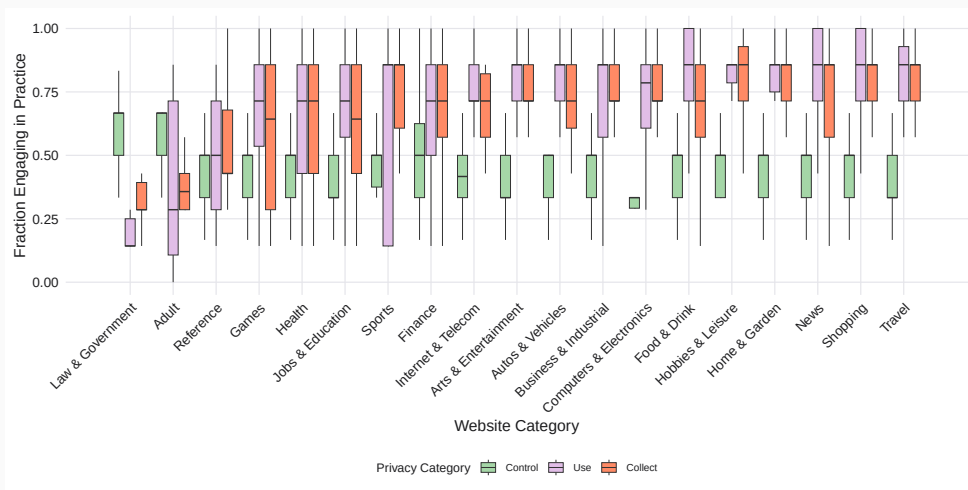
What goes into the Privacy term — and how do we measure it?

We read each site's privacy policy into a **19-attribute** space (top 500 sites):

- **Collection** — what data a site collects
- **Use** — how it uses or shares the data
- **Control** — how much control users have

Using this theoretical framework, we empirically measure actual practices, beliefs, and valuations.

Do privacy practices vary enough to matter?

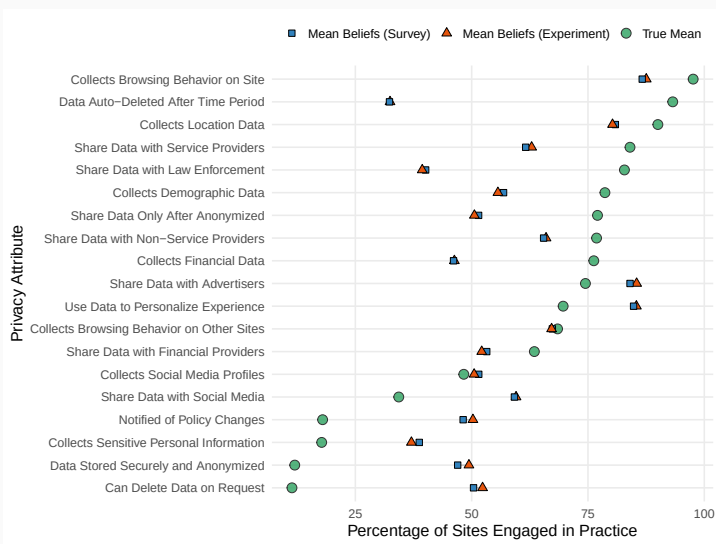


→ Websites in the same category handle data differently. Better information might change which website users choose.

Do consumers know and value these practices?

Now that we know actual practices, we measure consumers' beliefs and valuations.

What consumers believe vs. the truth



→ Consumers are systematically misinformed; beliefs are far from the truth.

Which privacy facts are decision-relevant?

Collection - your sensitive personal data (religion, political beliefs, sexuality, etc.)	Yes	No	Yes	Yes
Usage - your data is shared with social media	No	No	Yes	Yes
Usage - your data is shared with financial providers	No	Yes	Yes	Yes
Usage - your data is shared with advertisers and used for targeted advertising	No	Yes	Yes	Yes
Usage - data is used to personalize your experience	Yes	Yes	Yes	No
Control - your data is automatically deleted after a time period	No	Yes	Yes	No
Price	\$3.00	\$4.00	\$5.00	\$1.00
	Select	Select	Select	Select

NONE: I wouldn't choose any of these.

Select

→ Conjoint choices recover dollar-valued WTP for each privacy attribute.

Recovering user-level privacy preferences

Challenge: 12 tasks to estimate 19 attributes + price per user.

Hierarchical Bayes framework

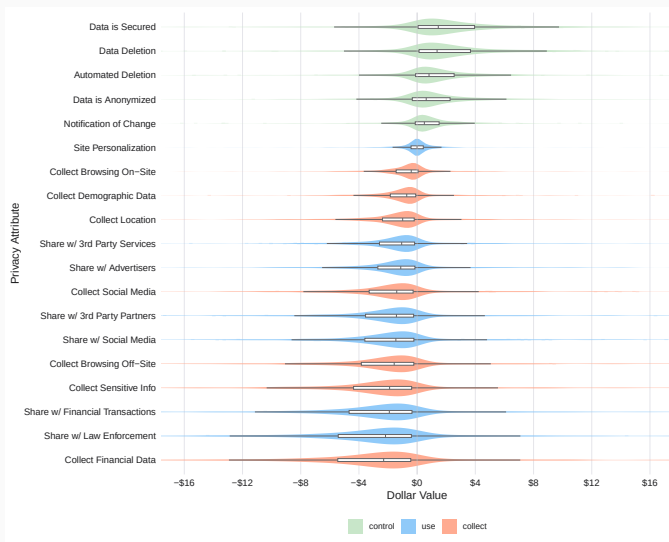
$$\underbrace{\beta_i}_{\text{Individual}} = \underbrace{\mu}_{\text{Population}} + \underbrace{\eta_i}_{\text{Heterogeneity}}, \quad \eta_i \sim \mathcal{N}(0, \Sigma)$$

Econometric advantages of hierarchical Bayes

- **Complete pooling** estimates only average WTP.
- **No pooling** is too noisy with 12 tasks for 19 attributes plus price.
- **Partial pooling** borrows information across users, giving stable user-level WTP while preserving heterogeneity.

→ Stable user-level valuations $\text{WTP}_{ik} = -\beta_{ik}/\beta_{i,\text{price}}$

What privacy attributes are worth



→ Consumers value privacy most when data use has off-platform consequences.

Privacy policies exist. Why are beliefs still wrong?

The experiment separates search costs from complexity.

Privacy policies are hard to find and read

Privacy policies are often hard to find, and difficult to read.

NYT privacy policy:

- **11,868 words**
- Average reading time: **17.8 min**

Two main frictions:

- **Search frictions:** finding and attending
- **Complexity:** internalizing policy content

The New York Times Company Privacy Policy

English (US)

Effective on May 27, 2026

At The New York Times Company, our mission is to seek the truth and help people understand the world. We want you to understand how we handle your personal information. We also want you to know your rights and choices.

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- Our websites, like nytimes.com and its subdomains including *The Athletic* and *Wirecutter*
- Our apps, like *The New York Times* app, *The New York Times Games* app, and *The New York Times Crossplay* app
- Our email newsletters, like *Cooking* and *Morning Briefing*
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If you have questions, please don't hesitate to contact us at privacy@nytimes.com. You can also review our [Privacy F.A.Q.](#)

Table of Contents

1. What Information Do We Gather About You?

1. (A) Information Collected through Times Services

1. Information You Voluntarily Give Us

• For Registration

• For Billing

• For User-Generated Content

• For Contests, Sweepstakes and Special Offers; [Notice of Financial Incentives](#)

• For Reader Surveys, Research, Panels and Experience Programs

• During Contact with Our Customer Care teams (including Call Centers (internal and Offshore), chat (including AI chatbots), and user reports)

• Your Contacts and Friends

• Refer-a-Friend

II. A Note about Sensitive Personal Information

Treatment 1 — Saliency

Q

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Saturday, May 17, 2025
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Long Drives and Short Homilies: How Father Bob Became Pope Leo

A résumé of deep religious education, pastoral experience, parish management, Vatican governance — and a nudge from Pope Francis — put Robert Prevost on the fast track.

15 MIN READ

A Century-Old Romance That Gave the Pope His Family Name

3 MIN READ



Gianni Cipriano for The New York Times



Sara Krulwich/The New York Times

Tonys 2025 Predictions: Who Will Win? And Who Should?

Our chief theater critic looks at this year's nominees and weighs in on the plays, musicals and artists he thinks will take home trophies.

A link to the website's privacy policy is provided.

Mechanism: Lower search costs

Treatment 2 — Information

Search icon

U.S. INTERNATIONAL CANADA ESPAÑOL 中文

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Via John Joseph Prevost



Sara Krulic/The New York Times

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Privacy Practices for this Website

Automated Deletion: **Yes** Collecting Demographic Data: **Yes**

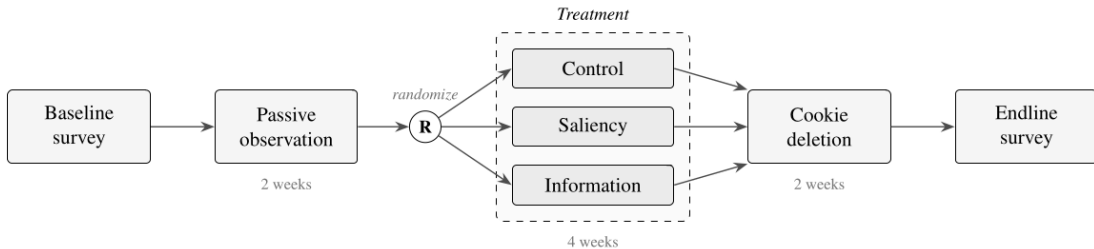
Site Personalization: **No**

[What do these fields mean?](#) [Show me all privacy attributes](#)

A condensed, personalized privacy summary is provided.

Mechanism: Lower search costs + lower complexity

Experimental timeline



End-of-intervention sample: Information **501**, Saliency **500**, Control **509**

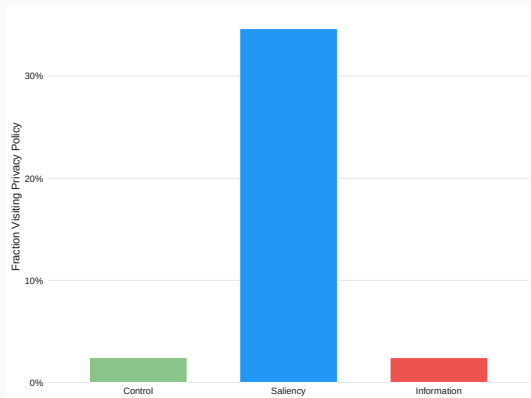
H1: Reducing information frictions should improve knowledge

If complexity is the binding friction, information should improve belief accuracy more than saliency.

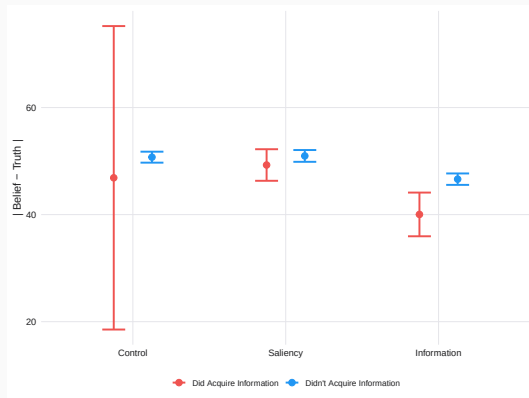
Model:	Belief – Truth (1)	Correctness (weak) (2)	Correctness (strict) (3)
Information Treatment	–4.971*** (0.8354)	0.0667*** (0.0137)	0.0715*** (0.0116)
Saliency Treatment	–0.3060 (0.8241)	–0.0098 (0.0139)	0.0076 (0.0120)
<i>Fixed-effects</i>			
Website FE	Yes	Yes	Yes
Block FE	Yes	Yes	Yes
<i>Clustered (experiment_id) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

→ Information improves accuracy: 5pp closer to truth; 6.7–7.2pp more likely correct.

H2: Search and learning are distinct



Policy visits: Control **2%**, Saliency **34%**



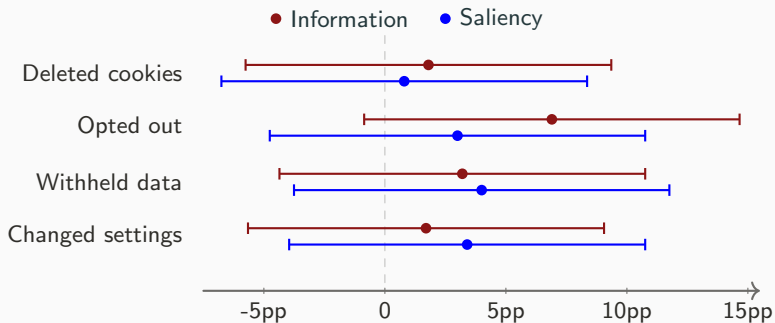
Search alone does not deliver learning.

→ Saliency raises policy visits $17\times$, but search alone does not produce learning.

Do better beliefs change behavior?

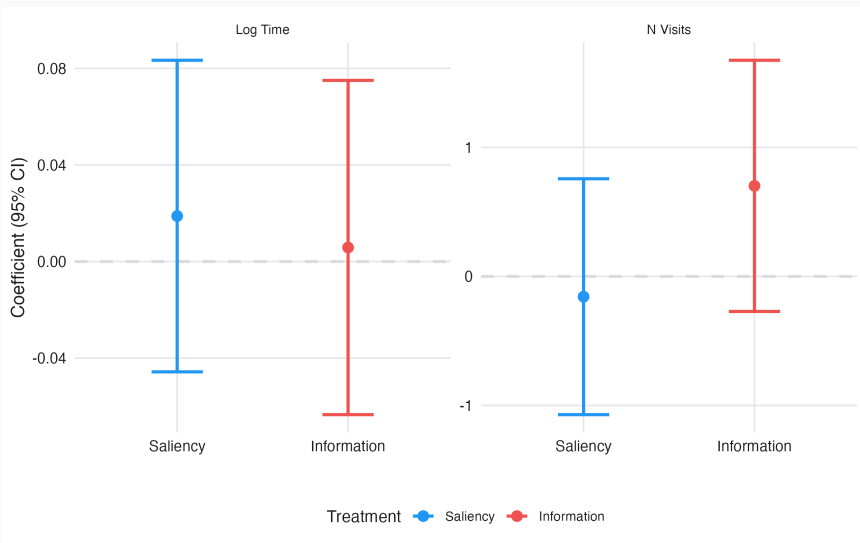
We test data sharing and website choice after the intervention.

H3: Self-reported data-sharing responses



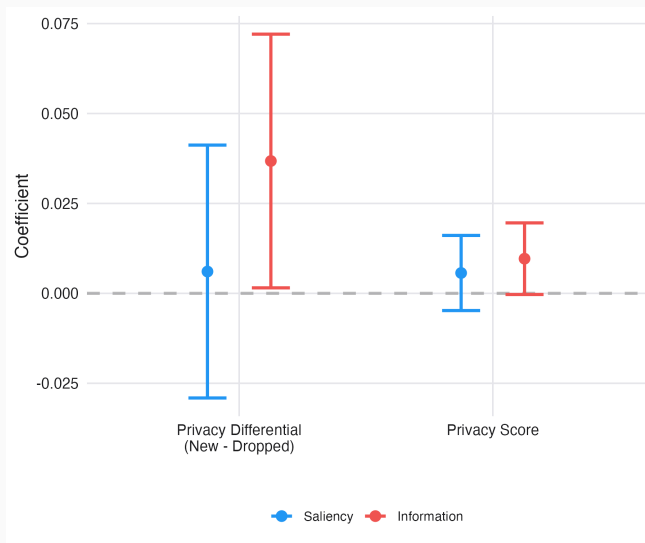
→ Information increases reported opt-outs by **6.9pp**.

H4: Existing-site usage



→ Core website usage does not change.

H4: Marginal site substitution



→ Newly added sites are modestly more private than dropped sites.

Conclusion

- **Access alone changes search, not learning**

Saliency raises policy visits $17\times$, but does not improve knowledge.

- **Simplified disclosure changes beliefs**

Information makes beliefs 5pp closer to truth and 6.7–7.2pp more likely correct.

- **Median WTP ranges from near \$0 to about \$2**

financial data, law-enforcement sharing, data security/deletion, and sensitive information rank highest.

- **Behavior changes are modest**

Users do not shift core website usage; adjustment appears mainly through marginal site substitution.

Thank you

markli@stanford.edu

Recruitment and baseline survey

Participants are recruited from **CloudResearch Connect** and **Prolific**, advertised generically as a study about online behavior.

The baseline survey gathers:

- beliefs about website privacy attributes
- stated preferences over privacy attributes
- willingness-to-accept (WTA) for installing the extension

Extension invitation depends on the incentive-compatible WTA draw.

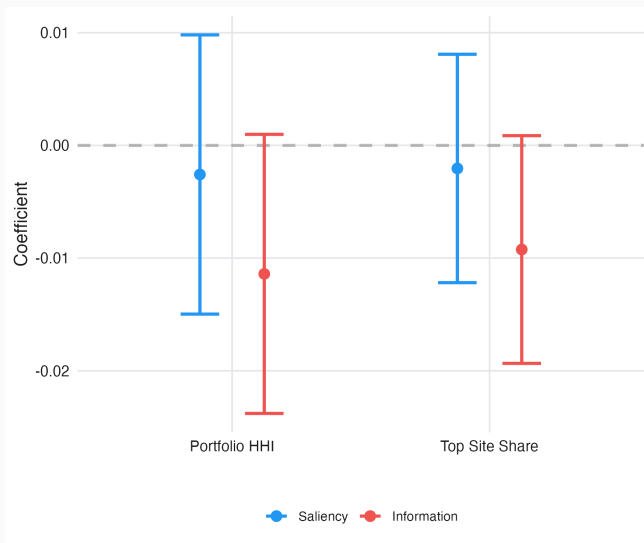
Challenge #1: Selection on privacy concerns

- **First: selection into platform** → Comparable to nationally representative PEW survey plots
- **Second: selection into the survey** → **randomized survey payments** → Survey payments should not drive differential selection
- **Third: selection into the observation** → **IC mechanism** → Which part of privacy distribution are we studying?

Sample	WTA for extension
Extension sample	median \$20, mean \$19
Overall survey sample	median \$25, mean \$52

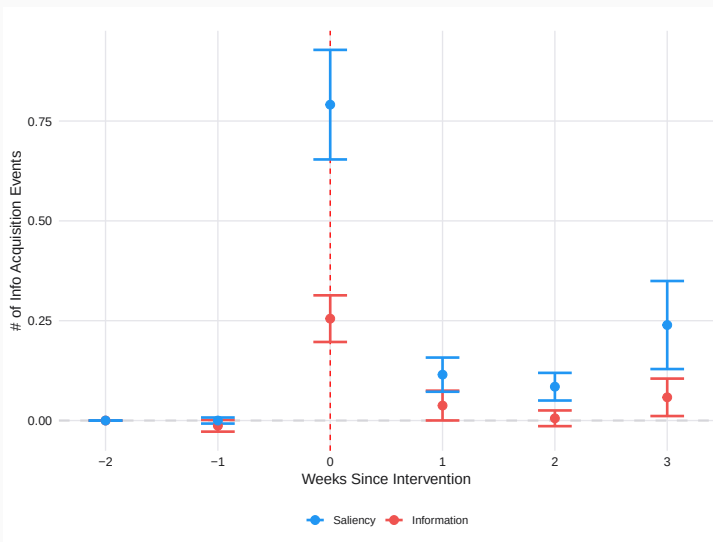
→ The \$40 cap cuts off the top **11%** of participants with $WTA > 40$.

Browsing set and concentration are stable



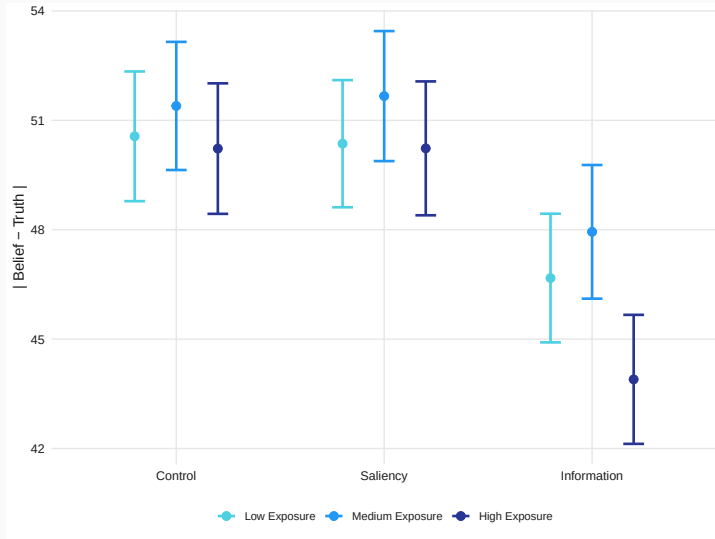
→ The privacy-ward extensive-margin result is not driven by broad disruption of browsing concentration.

Search over time



→ Search increases sharply when the intervention begins and remains elevated during treatment.

Learning across ex-ante exposure



→ Information improves belief accuracy across prior exposure terciles.

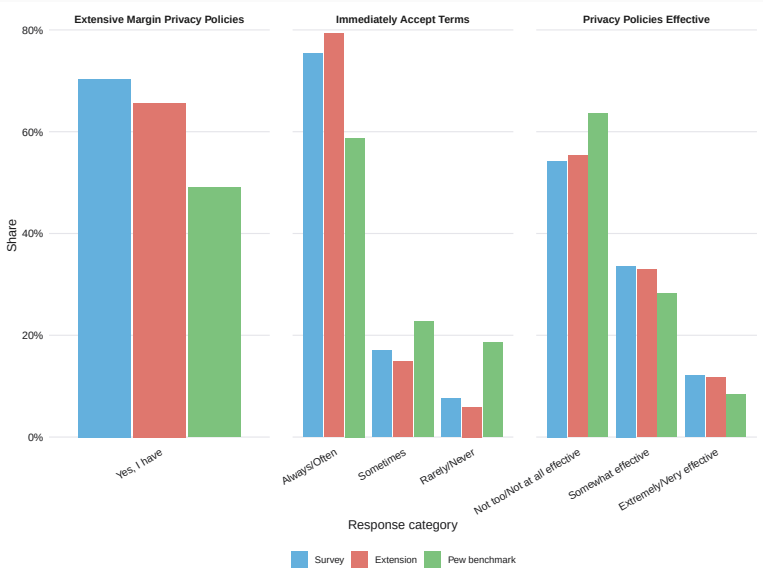
Correctness for non-provided information

Same test, for a random non-provided attribute:

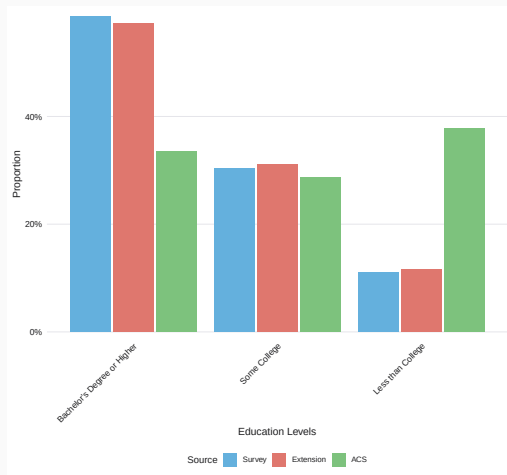
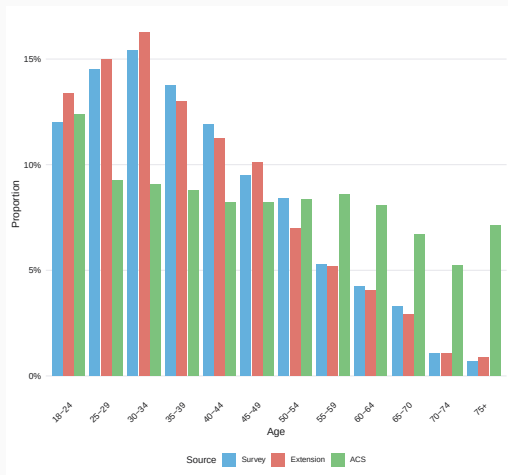
	Belief – Truth	Correct (weak)	Correct (strict)
Information	–2.67*	+0.040*	+0.042**
Saliency	+2.07	–0.045**	–0.029

→ Learning is strongest for attributes we actually disclose.

Comparison to PEW

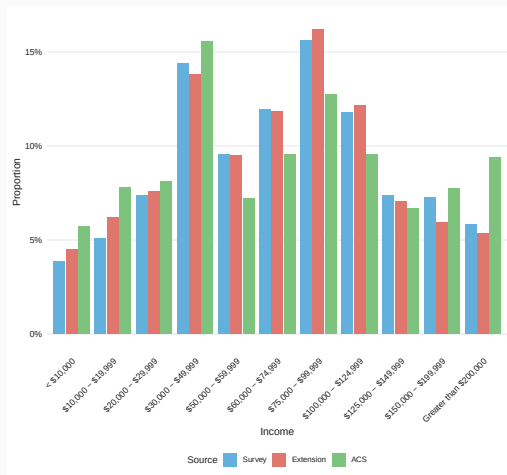
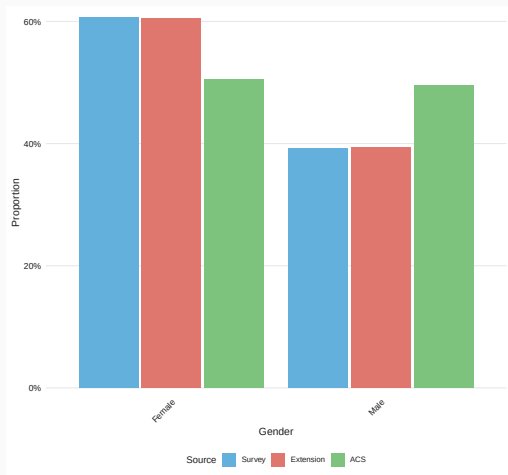


Demographic representativeness: age and education



→ The sample is reasonably representative but skews younger and more educated.

Demographic representativeness: gender and income



→ The sample is reasonably representative but skews female and more educated.

Hierarchical Bayes: priors and full spec

$$\beta_n = \mu_{c(n)} + \mathbf{L}_{\Sigma_{c(n)}} \mathbf{z}_n, \quad \mathbf{z}_n \sim \mathcal{N}(0, I) \quad (\text{non-centered})$$

$$\mu_c \sim \mathcal{N}(0, 5^2 I), \quad \tau_{c,f} \sim \text{Lognormal}(0, \sigma_\tau^2), \quad \mathbf{L}_\Omega \sim \text{LKJcorr}(5)$$

Website random intercept (sum-to-zero) + demographic utility shifter.

NUTS via `reduce_sum`; 4 chains, 2000 warmup / 3000 sampling.